# 🏦 Lab 7 – DataVault Bank
## Veritabanı Güvenliği ve Denetimi Deneyimi
### Senaryo
Bir bilgi sistemleri denetçisi olarak DataVault Bank'ın kritik müşteri verilerini barındıran Oracle veritabanını denetlemek üzere görevlendirildiniz.
Veritabanı yöneticileri aşağıdaki güvenceleri vermektedir:

- Yetkiler minimum seviyededir.
- Hassas veriler korunmaktadır.
- Şifreleme kullanılmaktadır.
- Ayrıcalıklı erişimler gözden geçirilmektedir.
- Veritabanı günceldir.
- Loglama etkin durumdadır.
- Görevler ayrılığı uygulanmaktadır.
- Varsayılan hesaplar kapatılmıştır.

Yalnızca aşağıdaki çıktılar sizinle paylaşılmıştır.
Amacınız riskleri belirlemek ve denetim raporu hazırlamaktır.

### DATABASE OVERVIEW

```
Database Name : DATAVAULTPRD
Version       : Oracle Database 19c Enterprise Edition
Environment   : Production
Size          : 2.3 TB
Records       : 42,318,441
Owner         : BANKDBA
```

### DATABASE USERS

```
SELECT username,
       account_status,
       profile,
       created
FROM dba_users;
```

| USERNAME | STATUS | PROFILE | CREATED |
|---|---|---|---|
| SYS | OPEN | DEFAULT | 2019 |
| SYSTEM | OPEN | DEFAULT | 2019 |
| DBSNMP | OPEN | DEFAULT | 2019 |
| APPUSER | OPEN | APP_PROFILE | 2020 |
| REPORTUSER | OPEN | DEFAULT | 2020 |
| TESTUSER | OPEN | DEFAULT | 2021 |
| LEGACYAPP | OPEN | DEFAULT | 2018 |

| | | | |
|---|---|---|---|
| HRREADONLY | OPEN | DEFAULT | 2020 |
| DEVTEAM | OPEN | DEFAULT | 2022 |
| TEMPCONSULTANT | OPEN | DEFAULT | 2023 |

**LAST LOGIN INFORMATION**

| USERNAME | LAST LOGIN |
|---|---|
| SYS | 2026-06-01 |
| SYSTEM | 2026-06-02 |
| TESTUSER | 2024-09-11 |
| TEMPCONSULTANT | 2025-02-03 |
| LEGACYAPP | 2023-11-12 |

**DATABASE ROLES**

```
SELECT grantee, granted_role
FROM dba_role_privs;
```

| USER | ROLE |
|---|---|
| SYS | DBA |
| SYSTEM | DBA |
| APPUSER | DBA |
| REPORTUSER | DBA |
| DEVTEAM | DBA |
| TESTUSER | CONNECT |
| HRREADONLY | CONNECT |
| TEMPCONSULTANT | DBA |

**SYSTEM PRIVILEGES**

```
SELECT grantee, privilege
FROM dba_sys_privs;
```

| USER | PRIVILEGE |
|---|---|
| APPUSER | CREATE USER |
| APPUSER | DROP USER |
| APPUSER | ALTER SYSTEM |
| DEVTEAM | CREATE TABLE |
| DEVTEAM | CREATE PROCEDURE |
| DEVTEAM | ALTER SESSION |
| REPORTUSER | SELECT ANY TABLE |
| REPORTUSER | EXECUTE ANY PROCEDURE |

## PASSWORD POLICY

```
SELECT profile,
       resource_name,
       limit
FROM dba_profiles;
```

| PROFILE | RESOURCE | LIMIT |
|---|---|---|
| DEFAULT | PASSWORD_LIFE_TIME | UNLIMITED |
| DEFAULT | PASSWORD_REUSE_TIME | UNLIMITED |
| DEFAULT | FAILED_LOGIN_ATTEMPTS | UNLIMITED |
| DEFAULT | PASSWORD_LOCK_TIME | DEFAULT |
| DEFAULT | PASSWORD_VERIFY_FUNCTION | NULL |

## PASSWORD HASHES

```
SELECT username,password_versions
FROM dba_users;
```

| USER | PASSWORD_VERSION |
|---|---|
| SYS | 10G 11G 12C |
| SYSTEM | 10G 11G 12C |
| APPUSER | 10G 11G |
| REPORTUSER | 10G |
| DEVTEAM | 10G |
| LEGACYAPP | 10G |

## CRITICAL TABLES

```
SELECT owner,
       table_name
FROM dba_tables;
```

| OWNER | TABLE |
|---|---|
| COREBANK | CUSTOMERS |
| COREBANK | CREDIT_CARDS |
| COREBANK | CUSTOMER_ADDRESSES |
| COREBANK | CUSTOMER_ACCOUNTS |

| | |
|---|---|
| COREBANK | CUSTOMER_TRANSACTIONS |
| COREBANK | INTERNET_BANKING_USERS |
| COREBANK | EMPLOYEES |
| COREBANK | LOANS |

**SAMPLE DATA**

**CUSTOMERS**

| ID | NAME | TCNO |
|---|---|---|
| 1001 | Mehmet Kaya | 12345678901 |
| 1002 | Ahmet Demir | 98765432100 |

**CREDIT_CARDS**

| CARD_NO | EXPIRY |
|---|---|
| 4111111111111111 | 09/28 |
| 5555555555554444 | 10/29 |

**EMPLOYEES**

| ID | NAME | SALARY |
|---|---|---|
| 101 | Ayşe Kaya | 215000 |
| 102 | Mehmet Arslan | 187000 |

**ENCRYPTION STATUS**

```
SELECT tablespace_name,
       encrypted
FROM dba_tablespaces;
```

| TABLESPACE | ENCRYPTED |
|---|---|
| SYSTEM | NO |
| USERS | NO |
| COREBANK | NO |
| CUSTOMERDATA | NO |

**AUDIT CONFIGURATION**

```
SHOW PARAMETER audit;


audit_trail = NONE
```

**BACKUP CONFIGURATION**

```
RMAN CONFIGURATION
```

```
Retention Policy : NONE
Backup Encryption : OFF
Last Successful Backup : 2025-11-17
```

### PATCH LEVEL

```
SELECT version FROM v$instance;
```

```
Oracle Database 19.3.0.0
```

### DATABASE LINKS

```
SELECT db_link,
       username,
       host
FROM dba_db_links;
```

| DB_LINK | USERNAME |
|---|---|
| HRDB_LINK | HRADMIN |
| CRM_LINK | CRMADMIN |
| LEGACY_LINK | SYS |

### FAILED LOGINS

```
SELECT username,
       count(*)
FROM failed_logins
GROUP BY username;
```

| USERNAME | COUNT |
|---|---|
| SYS | 324 |
| SYSTEM | 119 |
| APPUSER | 68 |
| DEVTEAM | 42 |

### ACTIVE SESSIONS

```
SELECT username,
       machine
FROM v$session;
```

| USER | MACHINE |
|---|---|
| SYS | DBA-LAPTOP |
| APPUSER | APPSERVER01 |
| DEVTEAM | DEV-PC-14 |
| TEMPCONSULTANT | CONSULTANT-LAPTOP |

**CHANGE LOG**

| CHANGE ID | DESCRIPTION | APPROVAL |
|---|---|---|
| CHG-1001 | New User Creation | YES |
| CHG-1002 | Table Structure Change | YES |
| CHG-1003 | Privilege Modification | NO |
| CHG-1004 | Procedure Update | NO |